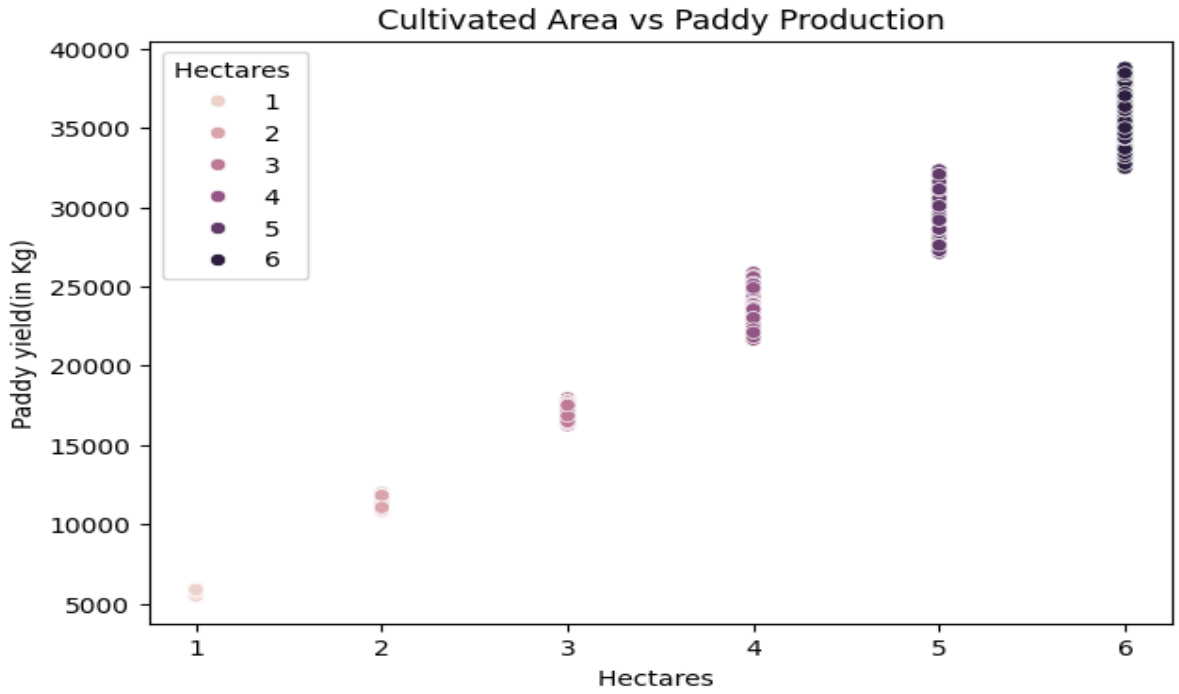
Paddy Crop Data Analysis and Prediction

Data Science Mini Project Report

Prepared by: Harishraj R

Program / Batch: Data Science • DS-ANB-04



A supervised machine-learning project using cultivation, weather, and farm-management features to estimate paddy yield in kilograms.

1. Project Overview

Agriculture is an important source of food production. This project explores paddy cultivation data to identify patterns and build supervised regression models that estimate **Paddy yield (in Kg)** from cultivation-related and environmental features.

Project objectives

• Explore dataset structure, descriptive statistics, missing values, duplicates, distributions, and feature relationships.

• Prepare data by removing duplicate rows and encoding categorical variables.

• Scale numeric features, select useful features, and split the data into training and test sets.

• Train and tune regression models using five-fold cross-validation.

• Compare model performance using MAE, RMSE, and R^2.

Project details

Item	Details
Project title	Paddy Crop Data Analysis and Prediction
Domain	Agriculture / Data Science
ML task	Supervised regression (continuous numeric prediction)
Target variable	Paddy yield(in Kg)
Original dataset	2,789 rows × 45 columns
After duplicate removal	2,338 rows × 45 columns
After one-hot encoding	2,338 rows × 64 columns
Train / test split	1,753 training rows / 585 test rows (75% / 25%)

Dataset source: Paddy dataset CSV on GitHub. The notebook loads the CSV from this URL, so an internet connection is required when running it.

2. Dataset and Methodology

The notebook reports 45 original columns, including numeric measures and categorical fields. Features include cultivated area, seed rate, fertilizer and pesticide inputs, rainfall and irrigation-related measurements, temperatures across growth stages, wind speed and direction, relative humidity, agricultural block, paddy variety, soil type, and nursery type.

Data preparation

• Missing-value inspection was performed; the displayed missing-value counts are zero for the original columns.

• Duplicate records were identified and removed. The row count changed from 2,789 to 2,338, meaning 451 duplicate rows were removed.

• Categorical columns were one-hot encoded, expanding the data from 45 to 64 columns.

• StandardScaler was applied to the numerical feature columns in the notebook. The target column was retained separately as the label for prediction.

• SelectKBest with an ANOVA F-test was used to select 32 features for model training.

• Data was split using train_test_split with test_size=0.25 and random_state=42.

Models and tuning

Three regression algorithms were compared: Decision Tree Regressor, Random Forest Regressor, and Gradient Boosting Regressor. Hyperparameters were tuned with RandomizedSearchCV using five-fold cross-validation. Model comparison used MAE, RMSE, and R^2 on the test set.

Tools and libraries

Python, NumPy, Pandas, Matplotlib, Seaborn, SciPy, and Scikit-learn. The provided requirements file also lists Jupyter and ipykernel.

Important validation note

The dataset contains many strongly correlated cultivation/input columns. Before presenting the model as a reliable real-world forecasting system, verify that derived or near-duplicate features do not leak the target or encode the same underlying quantity. The reported scores describe this notebook's current split and data; they are not proof of performance on unseen farms or future seasons.

3. Model Evaluation

The notebook’s five-fold cross-validation results and test-set evaluation are summarized below. Lower MAE/RMSE is better; R^2 closer to 1 indicates a stronger fit on the evaluated data.

Model	Best CV RMSE	Test MAE	Test RMSE	Test R^2
Decision Tree	820.352	669.559	928.552	0.989768
Gradient Boosting	821.000	663.897	920.300	0.989949
Random Forest	826.994	666.745	921.632	0.989920

Cross-validation selection: Decision Tree Regressor had the lowest best CV RMSE (820.352), so the notebook selected it as the best model based on cross-validation RMSE.

Test-set comparison: Gradient Boosting achieved the lowest test MAE (663.897), lowest test RMSE (920.300), and highest test R^2 (0.989949) among the three models. Thus, the model with the best cross-validation RMSE was not the model with the best test-set metrics.

Interpretation

The test-set R^2 values are approximately 0.99, indicating that the models explain a very high share of variation in the held-out data used here. However, the unusually high score should be investigated alongside the strong feature correlations and possible target leakage before claiming generalization. MAE and RMSE are in the target's unit, kilograms.

Project conclusion

This project demonstrates an end-to-end regression workflow: exploratory analysis, duplicate handling, categorical encoding, scaling, feature selection, train/test splitting, cross-validation, hyperparameter tuning, and metric-based model comparison. The next step is to validate the feature definitions and test the pipeline on independent data before deployment.

4. Exploratory Data Analysis Visuals

The following figures were supplied with the project and are included as visual evidence from the exploratory analysis. Interpretations are intentionally limited to what the plots show.

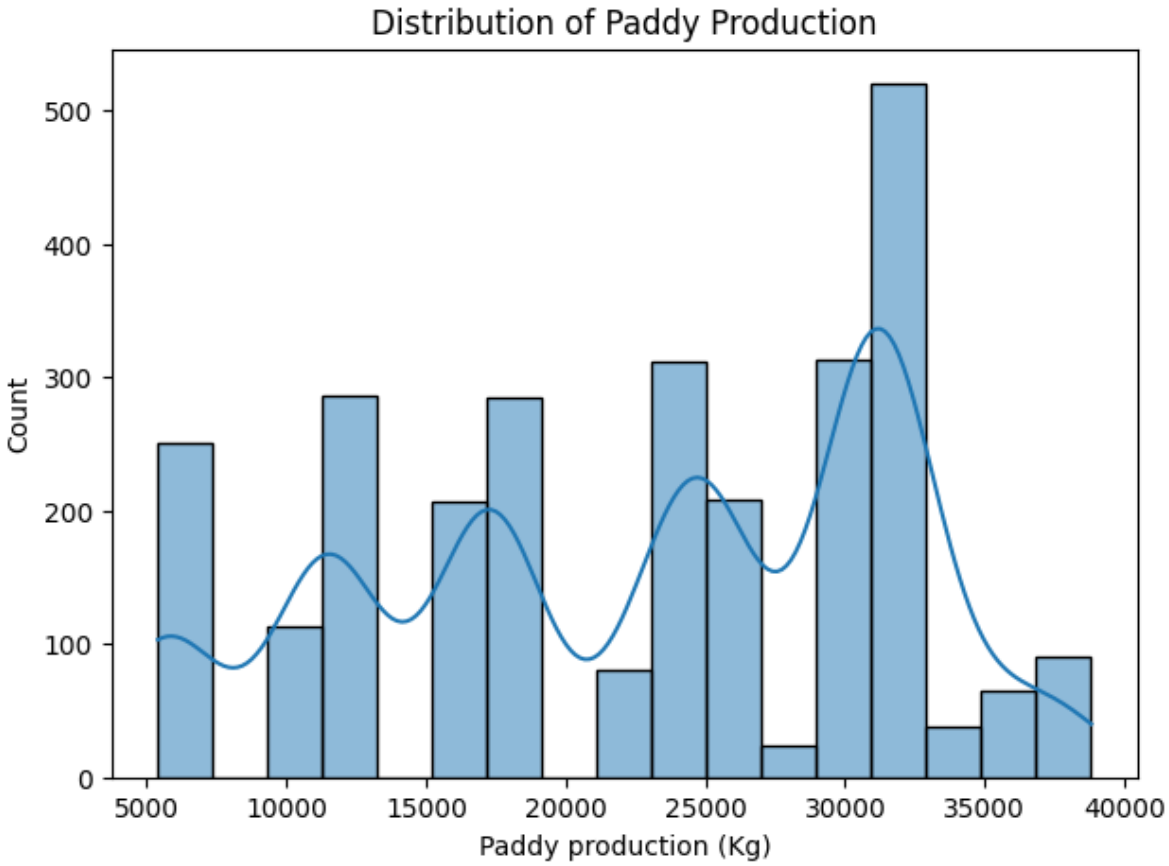


Figure 1. Distribution of paddy yield (kg). The plot shows multiple visible clusters across the production range.

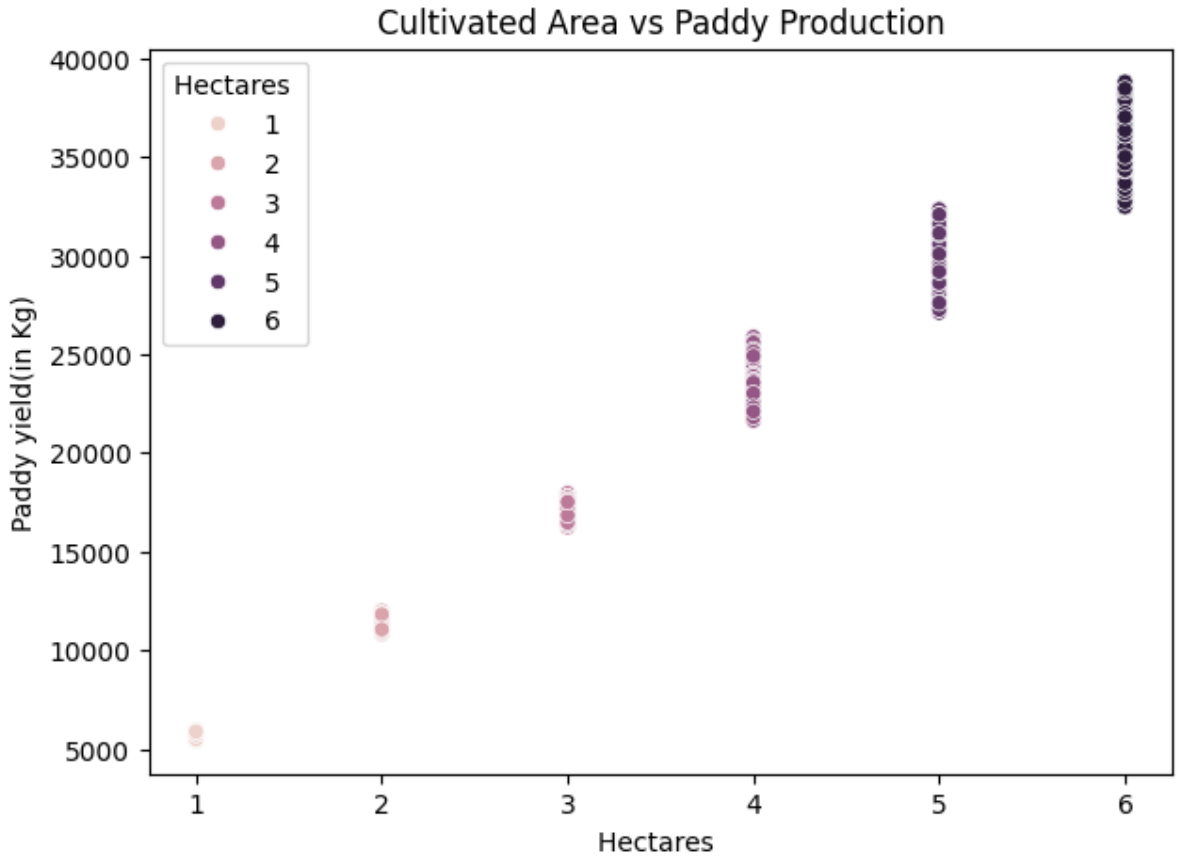


Figure 2. Cultivated area versus paddy production. Production rises in steps as recorded cultivated area increases; verify whether yield is directly derived from area or related inputs.

5. Relationships and Group Comparisons

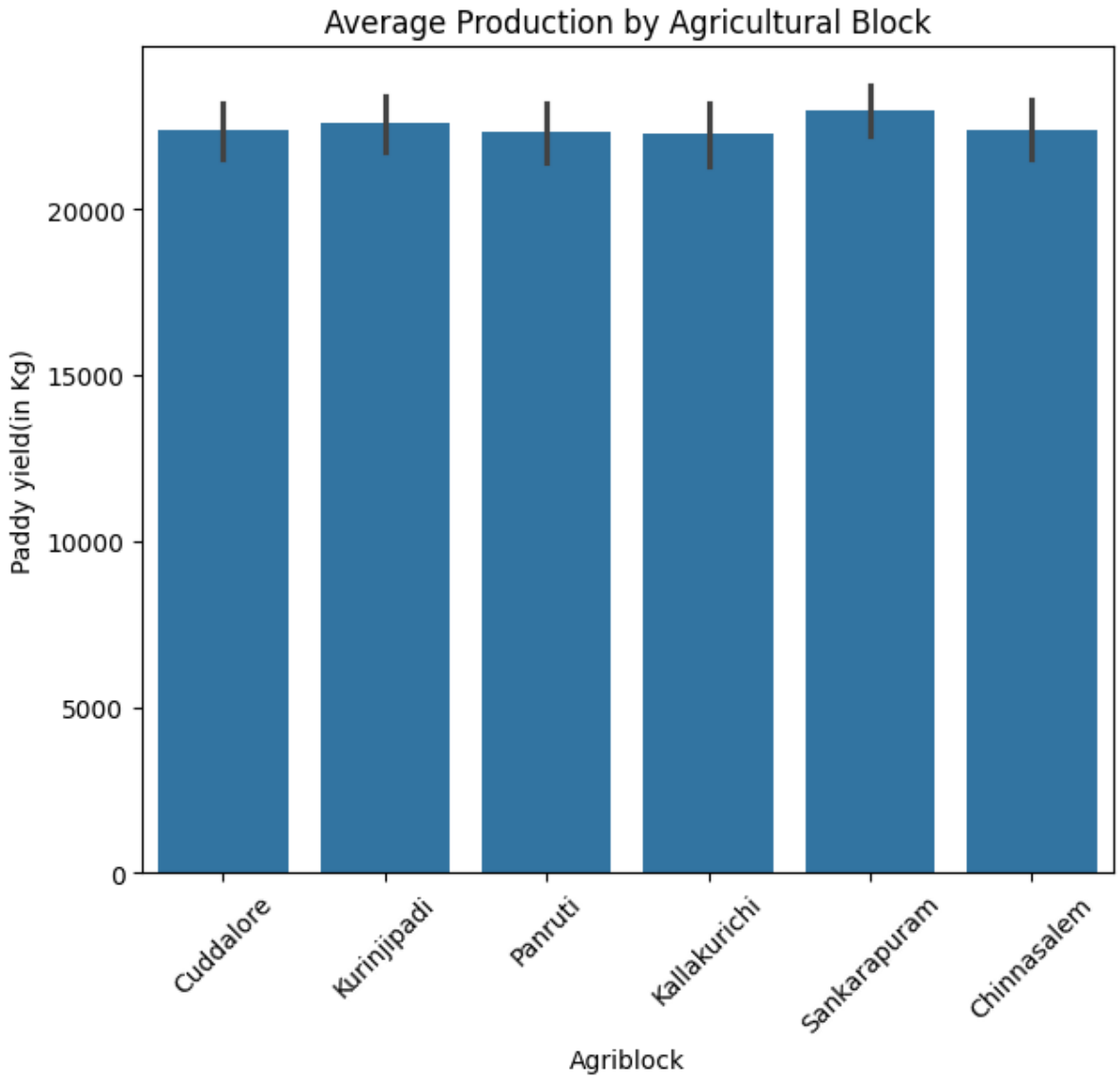


Figure 3. Average paddy production by agricultural block. The displayed means are relatively close, with Sankarapuram appearing somewhat higher.

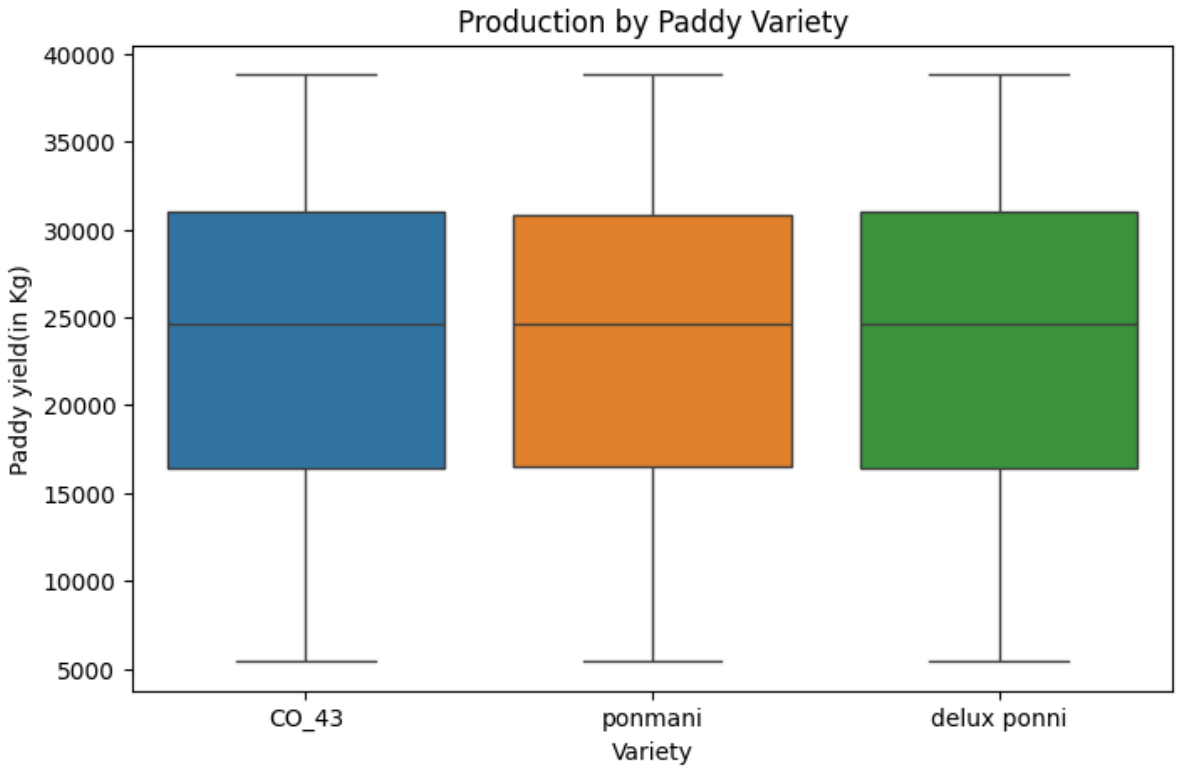


Figure 4. Paddy production by variety. The plotted distributions and medians appear very similar across the three varieties.

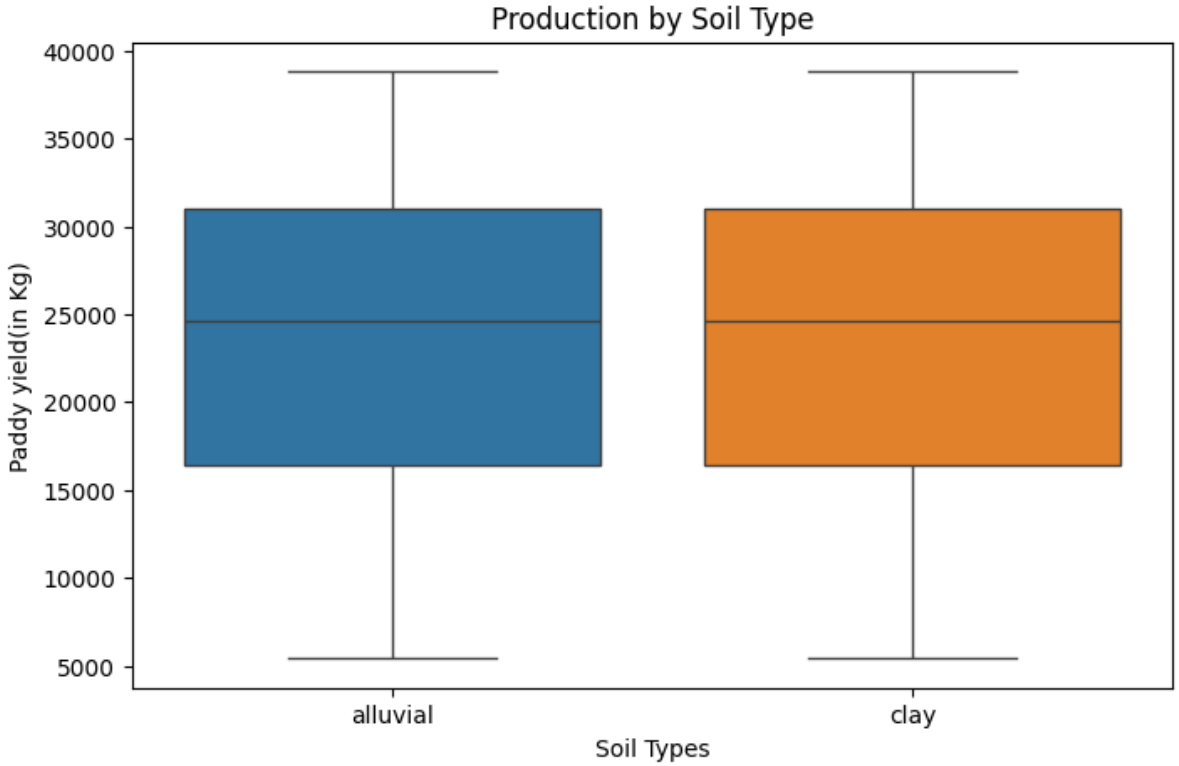


Figure 5. Paddy production by soil type. The two displayed distributions appear similar in their median and spread.

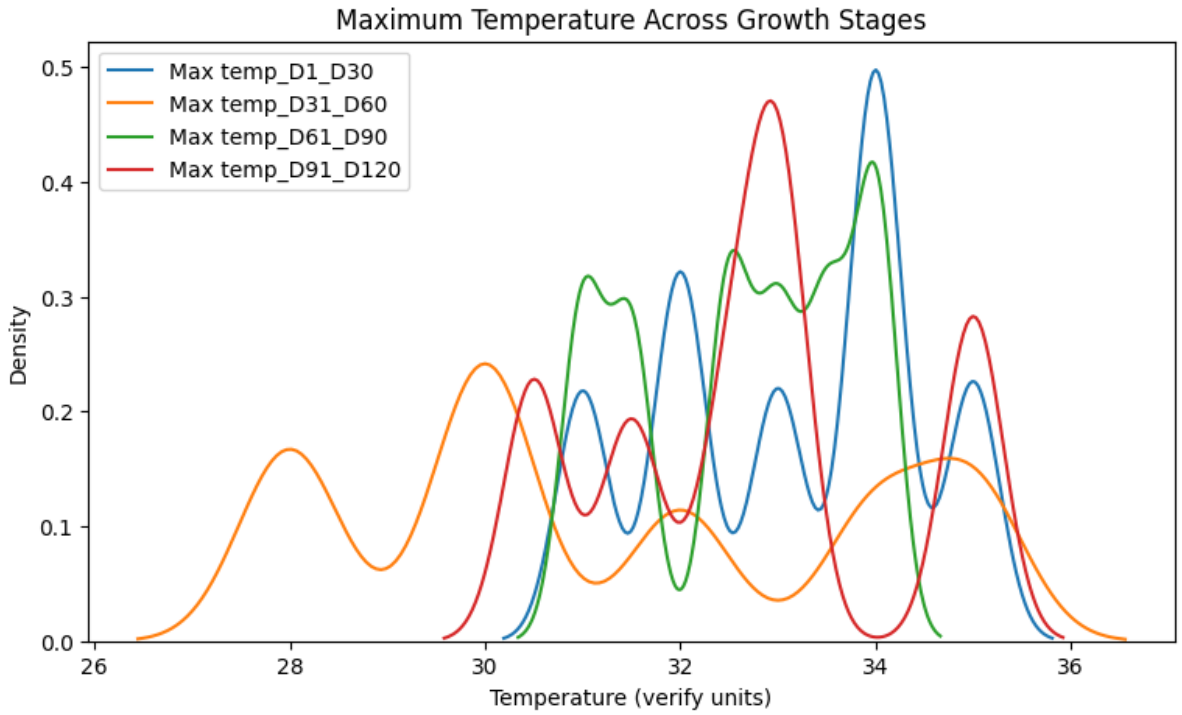


Figure 6. Maximum temperature distributions across four growth stages. The stage distributions differ, but confirm the temperature units and data definitions before interpretation.

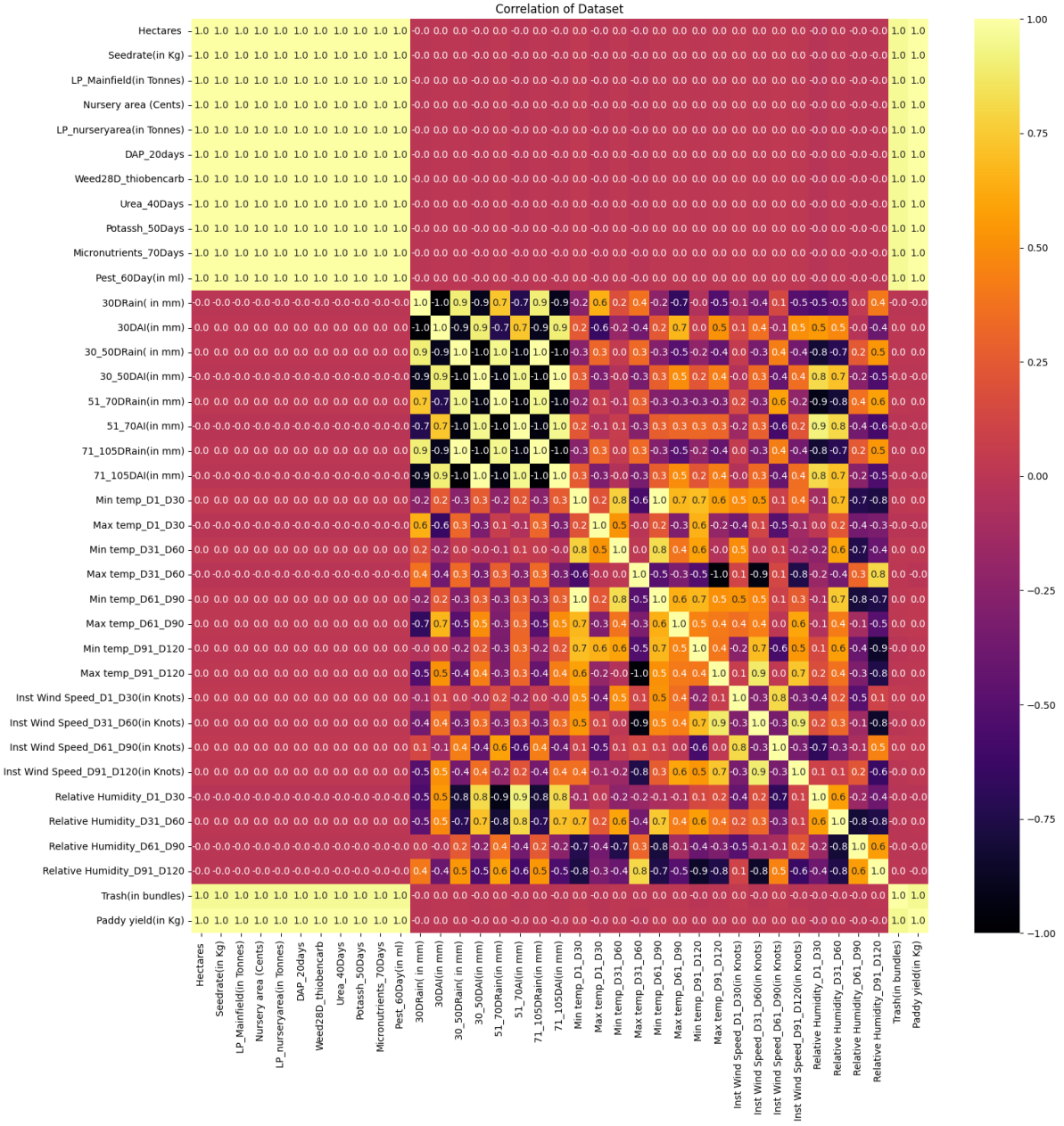


Figure 7. Correlation heatmap of numeric variables before categorical encoding. Several cultivation input variables show near-perfect correlations, which may reflect shared formulas or linked quantities.

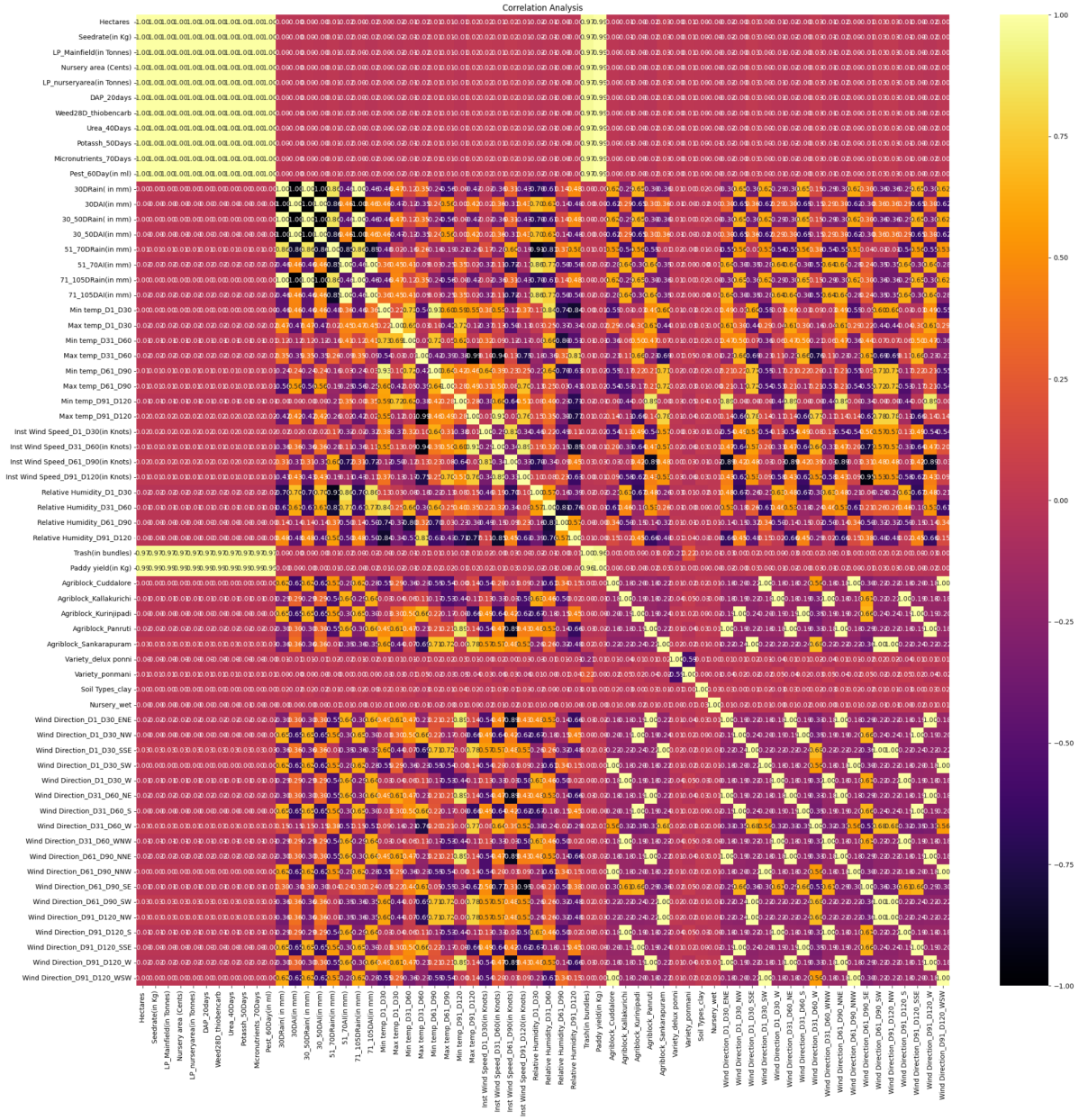


Figure 8. Correlation heatmap after encoding and feature engineering. The large matrix includes one-hot encoded categorical variables as well as numeric features.

6. Limitations and Next Steps

• Investigate the 451 duplicate rows: determine whether they are true duplicate records or repeated but valid observations before removing them permanently.

• Audit target leakage and redundant features, particularly where production, cultivated area, inputs, or derived quantities are mathematically linked.

• Use a preprocessing/model Pipeline fitted only on training folds. Feature selection and scaling should be fitted inside cross-validation to avoid data leakage.

• Validate on a separate season, farm, or geographic block to estimate real-world generalization.

• Keep units and feature definitions documented, especially temperature and rainfall fields.

• After validation, save the complete fitted preprocessing-plus-model pipeline and build an API/deployment workflow if required.

Project repository: Paddy Crop Data Analysis and Prediction on GitHub

Report prepared from the supplied Project_Notebook.ipynb, README, requirements file, and exported visualization images. Figures and reported metrics reflect the provided project artifacts; they have not been independently re-run for this report.